

Geometric mean

Example 1

If the weights of sorghum ear heads are 45, 60, 48, 100, 65 grams. Find the Geometric mean for the following data

Weight of ear head x (g)	Log x
45	1.653
60	1.778
48	1.681
100	2.000
65	1.813
Total	8.925

Solution

Here $n = 5$

$$\begin{aligned} \text{GM} &= \text{Antilog } \frac{\sum \log x_i}{n} \\ &= \text{Antilog } \frac{8.925}{5} \\ &= \text{Antilog } 1.785 \end{aligned}$$

Grouped Data

Example 12

Find the Geometric mean for the following

Weight of sorghum (x)	No. of ear head(f)
50	4
65	6
75	16
80	8
95	7
100	4

Solution

Weight of sorghum (x)	No. of ear head(f)	Log x	f x log x
50	5	1.699	8.495
63	10	10.799	17.99
65	5	1.813	9.065
130	15	2.114	31.71
135	15	2.130	31.95
Total	50	9.555	99.21

Here n= 50

$$GM = \text{Antilog} \left[\frac{\sum f \log x_i}{n} \right]$$

$$= \text{Antilog} \left[\frac{99.21}{50} \right]$$

$$= \text{Antilog } 1.9842 = 96.43$$

Example 13

For the frequency distribution of weights of sorghum ear-heads given in table below.

Calculate the Geometric mean

Weights of ear heads (in g)	No of ear heads (f)
60-80	22
80-100	38
100-120	45
120-140	35
140-160	20
Total	160

Solution

Weights of ear heads (in g)	No of ear heads (f)	Mid x	Log x	f log x
60-80	22	70	1.845	40.59
80-100	38	90	1.954	74.25
100-120	45	110	2.041	91.85
120-140	35	130	2.114	73.99
140-160	20	150	2.176	43.52
Total	160			324.2

Here $n = 160$

$$\begin{aligned}\text{GM} &= \text{Antilog} \left[\frac{\sum f \log x_i}{n} \right] \\ &= \text{Antilog} \left[\frac{324.2}{160} \right] \\ &= \text{Antilog} [2.02625] \\ &= 106.23\end{aligned}$$

Harmonic mean (H.M)

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocal of the given values. If $x_1, x_2, \dots, x_n$ are n observations,

$$\text{H.M} = \frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i} \right)}$$

For a frequency distribution

$$\text{H.M} = \frac{n}{\sum f \left(\frac{1}{x_i} \right)}$$

H.M is used when we are dealing with speed, rates, etc.

From the given data 5, 10, 17, 24, 30 calculate H.M.

X	$\frac{1}{x}$
5	0.2000
10	0.1000
17	0.0588
24	0.0417
30	0.4338

$$\text{H.M} = \frac{5}{0.4338} = 11.526$$

Example 14

Number of tomatoes per plant are given below. Calculate the harmonic mean.

Number of tomatoes per plant	20	21	22	23	24	25
Number of plants	4	2	7	1	3	1

Number of tomatoes per plant (x)	No of plants(f)	$\frac{1}{x}$	$f\left(\frac{1}{x}\right)$
20	4	0.0500	0.2000
21	2	0.0476	0.0952
22	7	0.0454	0.3178
23	1	0.0435	0.0435
24	3	0.0417	0.1251
25	1	0.0400	0.0400
	18		0.8216

$$H.M = \frac{n}{\sum f\left(\frac{1}{x_i}\right)} = \frac{18}{0.1968} = 21.91$$

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